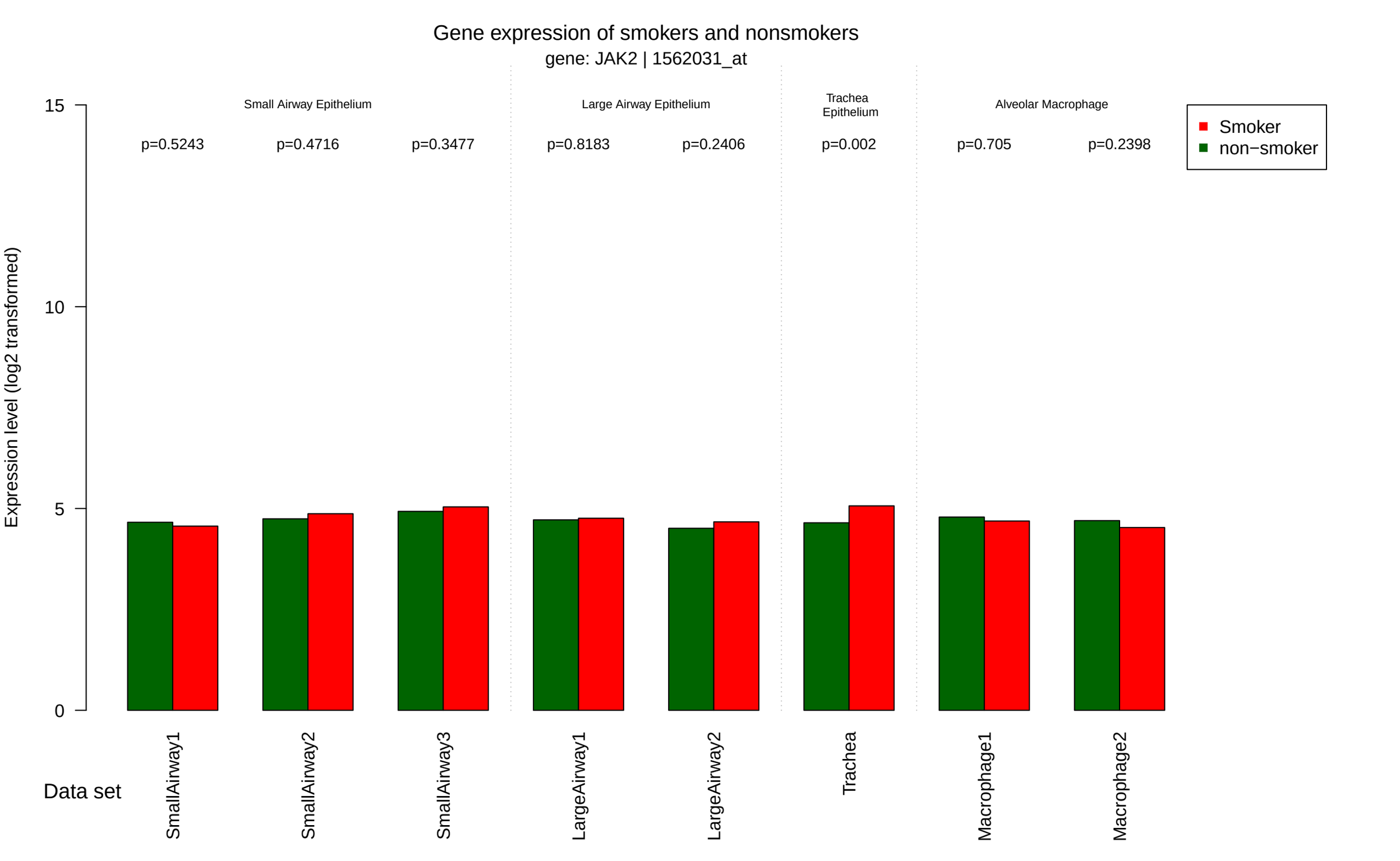
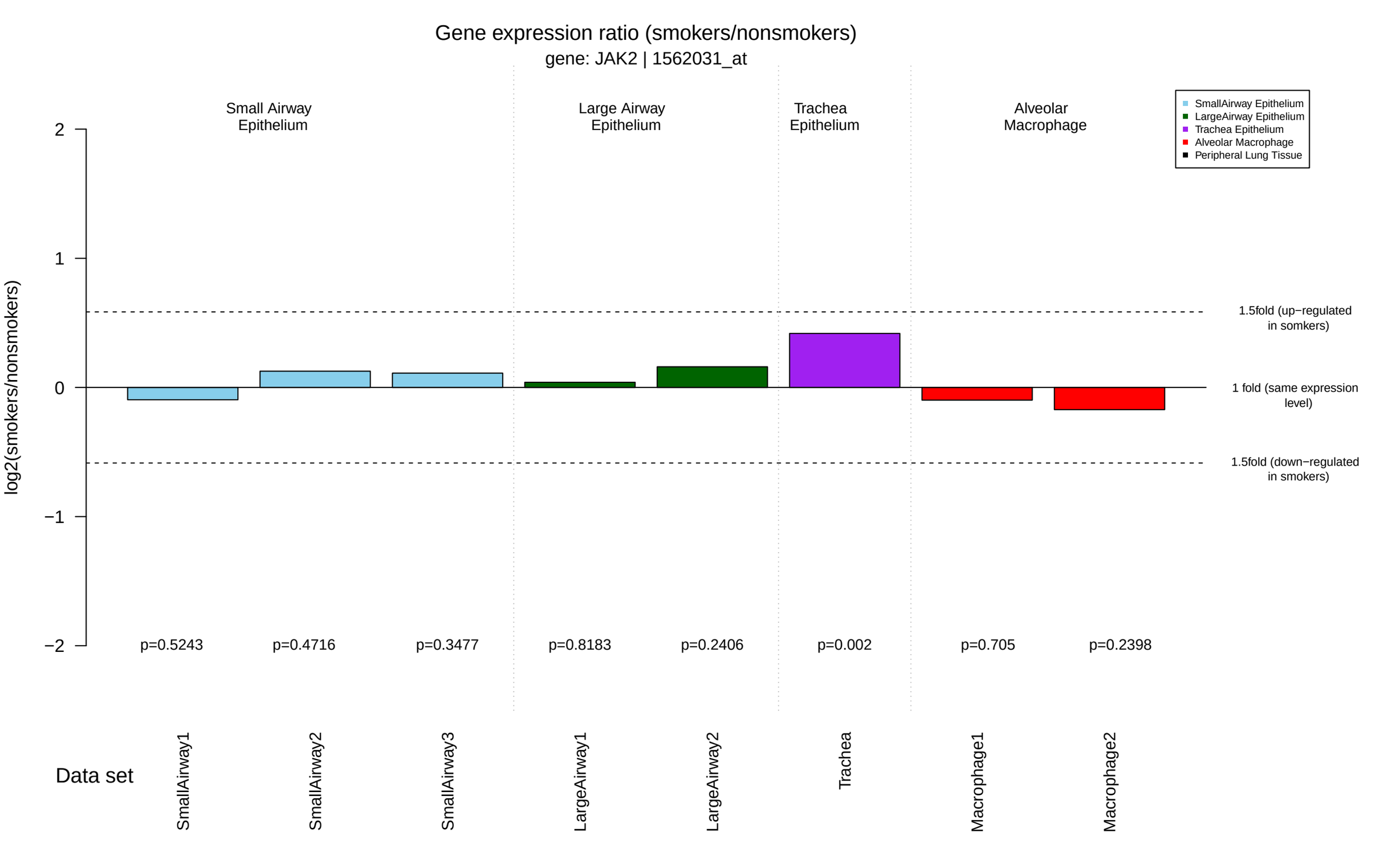
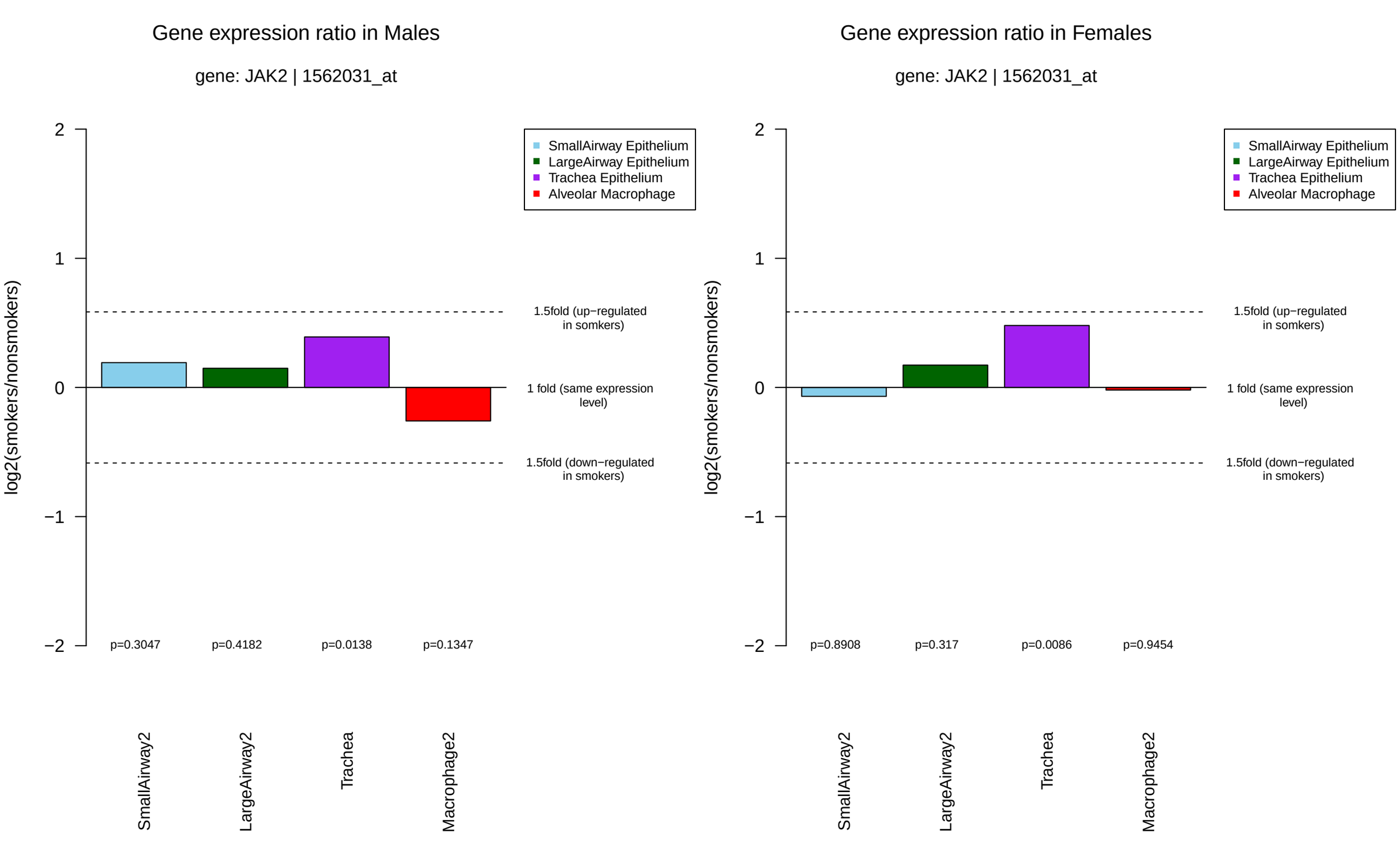
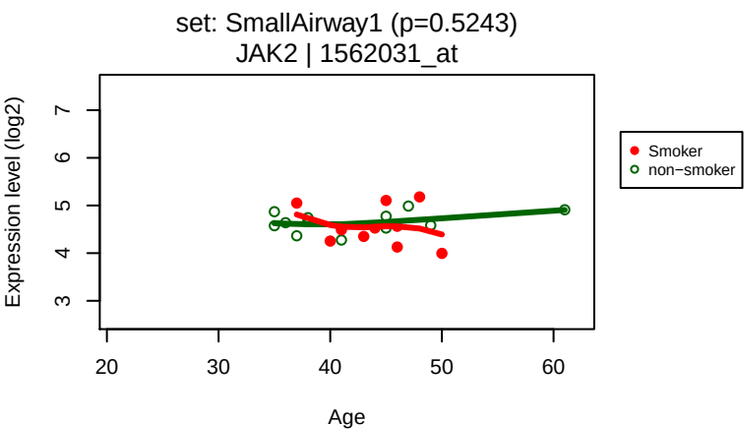
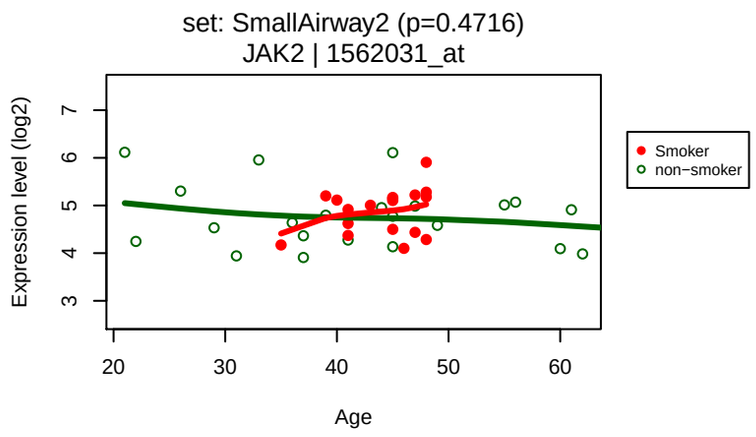
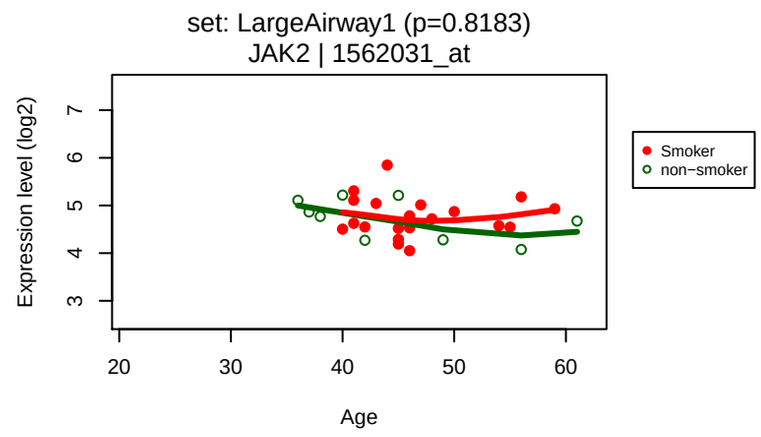

Gene expression of smokers and nonsmokers
gene: JAK2 | 1562031_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5243
p=0.4716
p=0.3477
p=0.8183
p=0.2406
p=0.002
p=0.705
p=0.2398
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

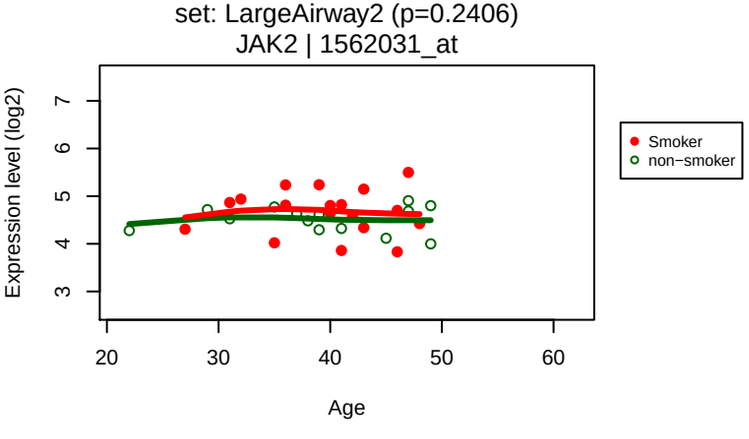

Gene expression ratio (smokers/nonsmokers)
gene: JAK2 | 1562031_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5243
p=0.4716
p=0.3477
p=0.8183
p=0.2406
p=0.002
p=0.705
p=0.2398
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

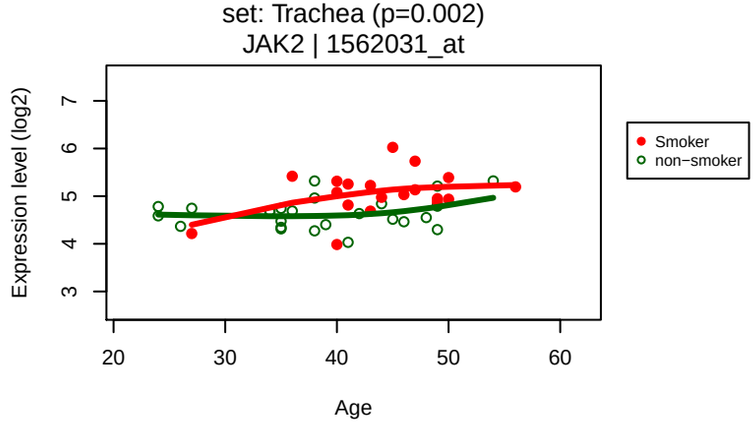

Gene expression ratio in Males
gene: JAK2 | 1562031_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3047
p=0.4182
p=0.0138
p=0.1347
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: JAK2 | 1562031_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8908
p=0.317
p=0.0086
p=0.9454
SmallAirway2
LargeAirway2
Trachea
Macrophage2

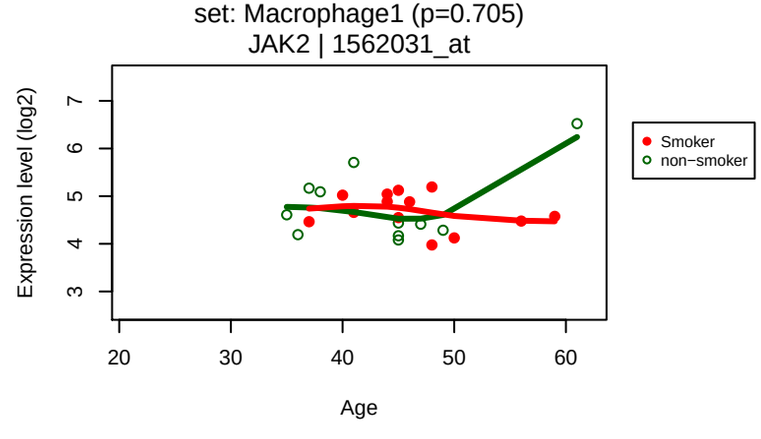

set: SmallAirway1 (p=0.5243)
JAK2 | 1562031_at
Expression level (log2)
Age
Smoker
non−smoker

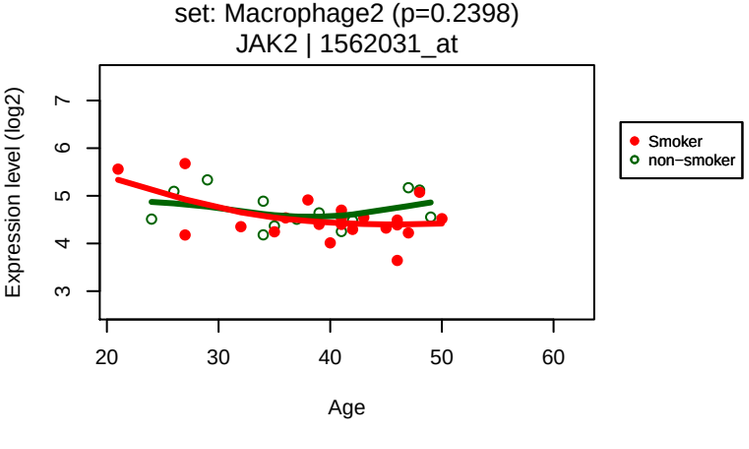

set: SmallAirway2 (p=0.4716)
JAK2 | 1562031_at
Expression level (log2)
Age
Smoker
non−smoker

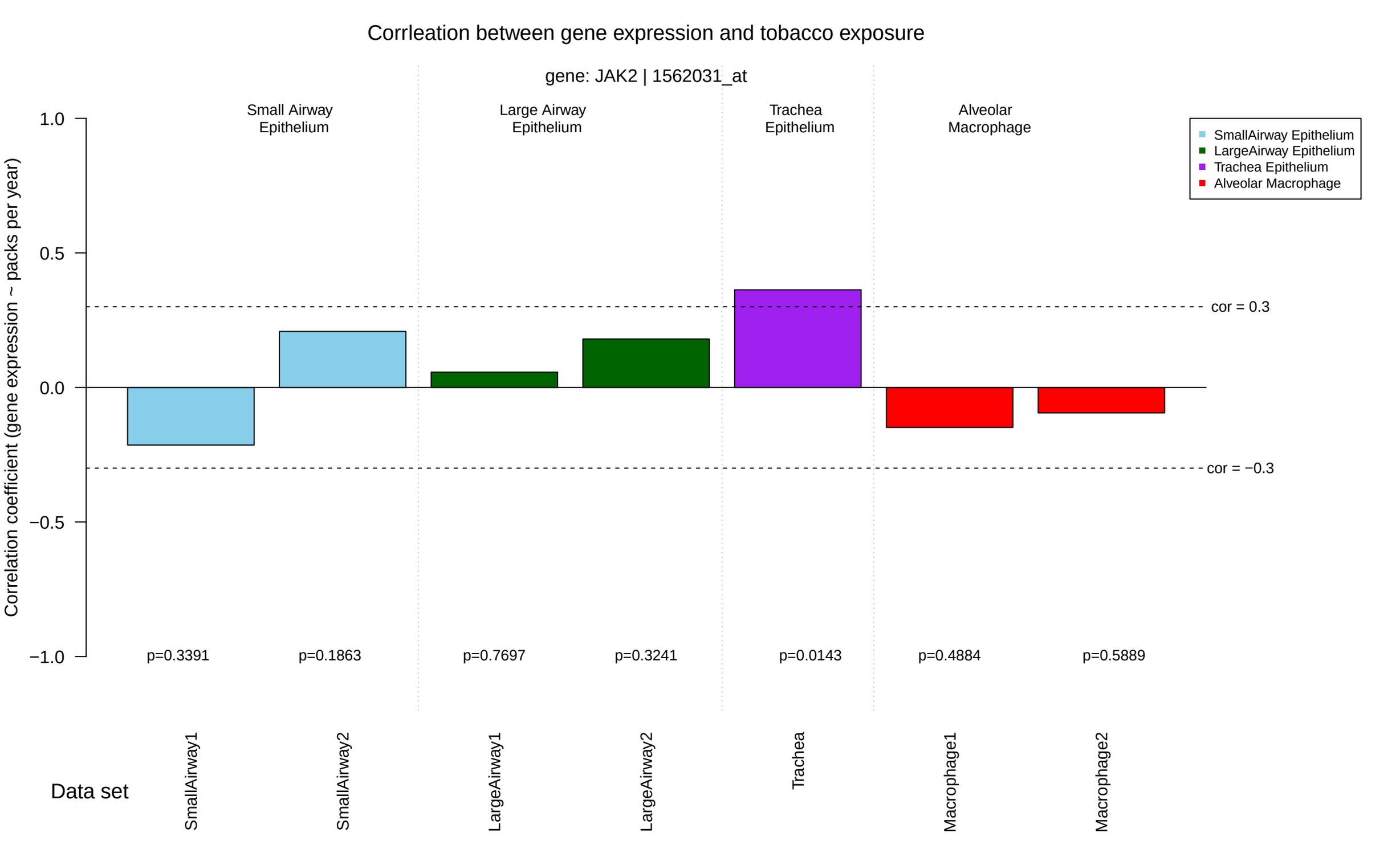

set: LargeAirway1 (p=0.8183)
JAK2 | 1562031_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2406)
JAK2 | 1562031_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.002)
JAK2 | 1562031_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.705)
JAK2 | 1562031_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2398)
JAK2 | 1562031_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: JAK2 | 1562031_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3391
p=0.1863
p=0.7697
p=0.3241
p=0.0143
p=0.4884
p=0.5889
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2